INNOVATIVE PRODUCT DEVELOPMENT REPORT

HELMET DETECTION USING CNN

Submitted by:

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In partial fulfillment of the Academic Requirements for the Degree of

BACHELOR OF TECHNOLOGY

Computer Science & Engineering



MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(Autonomous Institution-UGC, Govt. of India)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the work embodies in this dissertation entitled

"HELMET DETECTION"

being submitted by K.VAISHNAVI(23RH1AO5AE), K.SANJANA (23RH1A05CD), K.HARIKA (24RH5A0535) for partial fulfilment of requirement for for the award of BACHELOR OF TECHNOLOGY in Computer Science and Engineering discipline to Malla Reddy Engineering College for Women, Maisammaguda, Secunderabad during the academic year 2024-2025 is a record of bonafide piece of work, undertaken by her supervision of the undersigned.

Supervisor's Signature	**Head of the Department**
K.OBULESH	**Dr .Y.GEETHA REDDY**
Associate Professor	Professor and HOD

EXTERNAL EXAMINER

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DECLARATION

We hereby declare that our project entitled "**HELMET DETECTION**" submitted to Malla Reddy **Engineering College for Women, Hyderabad** for the award of the Degree of Bachelor of Technology in **Computer Science and Engineering** is a result of original research work done by us. It is declared that the project report or any part there of has not been previously submitted to any University or Institute for the award of Degree.

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We wish to convey gratitude to our Principal **Dr. Y. MADHAVEE LATHA**, for providing us with the environment and mean to enrich our skills and motivating us in our endeavor and helping us to realize our full potential.

We express our sincere gratitude to **Dr . Y.GEETHA REDDY**, Head of the Department of Computer Science and Engineering for inspiring us to take up a project to this subject and successfully guiding us towards its completion.

We would like to thank our Internal Guide **K.OBULESH**, Associate Professor and all the Faculty members for their valuable guidance and encouragement towards the completion of our project work.

With Regards and Gratitude,

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ABSTRACT

Helmet detection and license plate recognition are critical components of intelligent transportation systems (ITS) and public safety monitoring. This paper presents a Convolutional Neural Network (CNN)-based approach for real-time helmet detection and license plate recognition from vehicle images. The proposed method leverages the power of CNNs to automatically detect the presence of helmets on riders and extract license plate information from vehicles. The helmet detection model is trained on a large dataset of rider images, distinguishing between helmeted and non-helmeted riders with high accuracy. Similarly, the license plate recognition system employs CNNs to detect and classify license plates, overcoming challenges such as occlusions, different font styles, and environmental variations. The integrated solution provides a robust framework for enhancing road safety enforcement, monitoring traffic compliance, and automating security applications. Experimental results demonstrate the effectiveness of the system in various real-world conditions, achieving high detection accuracy, robust performance under varying lighting and angles, and low computational overhead for real-time applications. This work contributes to advancing AI-driven solutions for transportation safety and law enforcement.

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CHAPTER-I
INTRODUCTION

The application of deep learning, specifically Convolutional Neural Networks (CNNs), has revolutionized the field of computer vision, enabling accurate and efficient solutions for various real-world problems. Two such critical applications are helmet detection and license plate recognition, both of which serve vital functions in safety, security, and traffic management.

Helmet Detection: The safety of individuals in hazardous environments, such as construction sites, roads, or industrial settings, is a growing concern worldwide. Ensuring that workers or drivers are wearing helmets is crucial to reduce the risk of head injuries. Traditional methods of manual monitoring or simple image processing techniques are insufficient for real-time, large-scale deployment. CNNs, which are specifically designed for pattern recognition, can be trained to detect helmets in images or video streams with high accuracy, offering an automated solution for safety enforcement.

License Plate Recognition (LPR): License plate recognition is another key area where CNNs have made a significant impact. LPR systems are widely used in traffic monitoring, law enforcement, and parking management to automatically recognize and process vehicle license plates. The challenges associated with LPR include varying plate designs, lighting conditions, and occlusions, which can make traditional image processing techniques less effective. However, CNNs excel in learning complex visual patterns and can overcome these challenges by being trained on large datasets of vehicle images.

CHAPTER-II

TECHNOLOGIES LEARNT

2.1. EXISTING SYSTEM

Traditional helmet detection systems rely heavily on manual observation or basic image processing techniques such as color detection, edge detection, or background subtraction. These methods typically focus on detecting the color or shape of helmets in images or video streams. However, such techniques have significant limitations, especially in dynamic environments where lighting, camera angles, and occlusions can vary greatly.

1. Manual Monitoring: In many scenarios, helmet compliance is monitored by supervisors or security personnel. This approach is time-consuming, prone to human error, and inefficient in large-scale environments.

2. Image Processing-Based Methods: Early automated systems utilized conventional image processing techniques such as template matching or background subtraction. These methods often failed under complex conditions (e.g., low light, overlapping objects, or various helmet styles). They also required significant manual tuning of parameters, making them less adaptable to diverse environments.

3. Machine Learning-Based Approaches: Some existing systems employ classical machine learning techniques like Support Vector Machines (SVM) or decision trees, combined with hand-crafted features (e.g., Histogram of Oriented Gradients (HOG)). While these systems perform better than basic image processing, they are limited by the quality of feature extraction and are less accurate in detecting helmets in complex scenes.

These systems often face challenges in achieving high detection accuracy, handling large datasets, and adapting to varying environmental factors, such as different helmet colors or partially obscured views.

2.2 PROPOSED SYSTEM

The proposed system uses Convolutional Neural Networks (CNNs) to automatically detect helmets with higher accuracy and adaptability to complex environments. CNNs have revolutionized object detection by learning relevant features directly from data, eliminating the need for manual feature extraction.

1. Deep Learning-Based Detection: The proposed system employs a CNN architecture designed to detect helmets in images or video streams. The network is trained on a large and diverse dataset of individuals wearing and not wearing helmets. This enables the model to learn distinguishing features between helmets and other objects or headwear, resulting in more accurate detection, even in challenging conditions (e.g., varying lighting, different helmet styles, and complex backgrounds).

2. Real-Time Detection: The CNN-based model can be deployed in real-time surveillance systems, processing video frames on the fly to identify helmet use. This provides continuous monitoring of compliance in high-traffic environments such as construction sites or urban areas.

3. Improved Accuracy and Robustness: CNNs, with their ability to capture spatial hierarchies in images, significantly improve detection performance compared to traditional methods. The proposed system adapts to variations in camera angles, partial occlusions, and different environments without needing manual adjustments.

CHAPTER-III

SYSTEMREQUIREMENTS:

3.1. HARDWARE REQUIREMENTS:

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Floppy Drive : 1.44 Mb.
- Monitor : 15 VGA Colour.
- Mouse : Logitech.
- Ram : 512 Mb.

3.2. SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Coding Language: Python 3.7

CHAPTER-IV

SYSTEM DESIGN

System Design for Helmet Detection and License Plate Recognition Using CNN

To design a system for Helmet Detection and License Plate Recognition (LPR) using Convolutional Neural Networks (CNNs), the system can be divided into several components, each handling specific tasks. The overall system consists of image acquisition, image preprocessing, object detection, helmet detection, license plate extraction, and license plate recognition.

Below are the Block Diagram and UML Diagrams for this system:

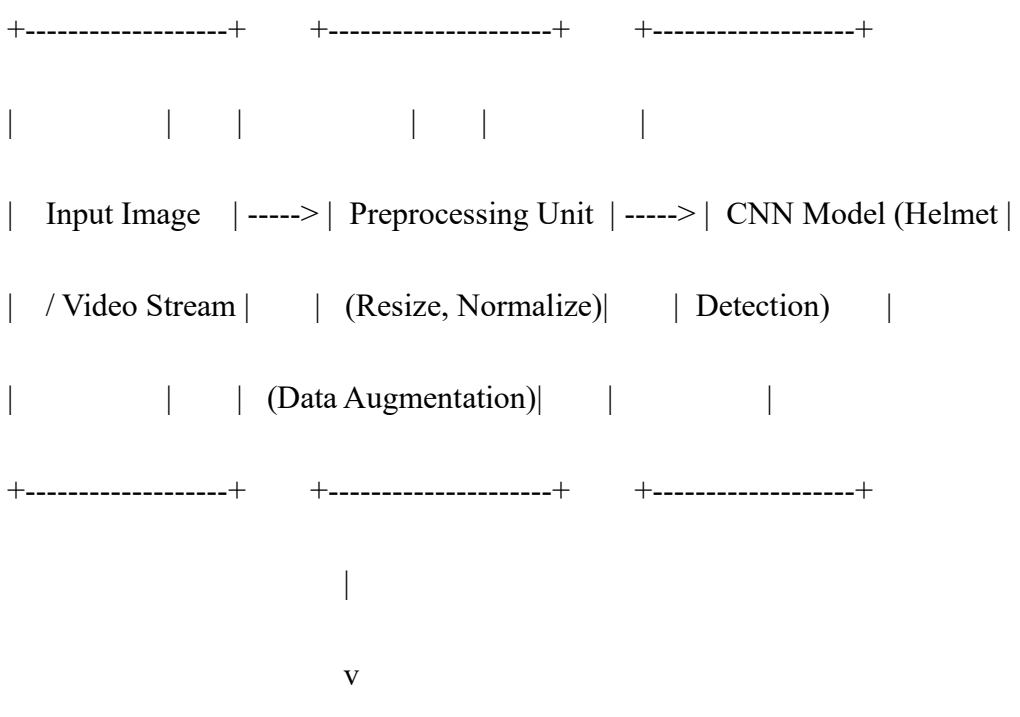
BLOCK DIAGRAM:

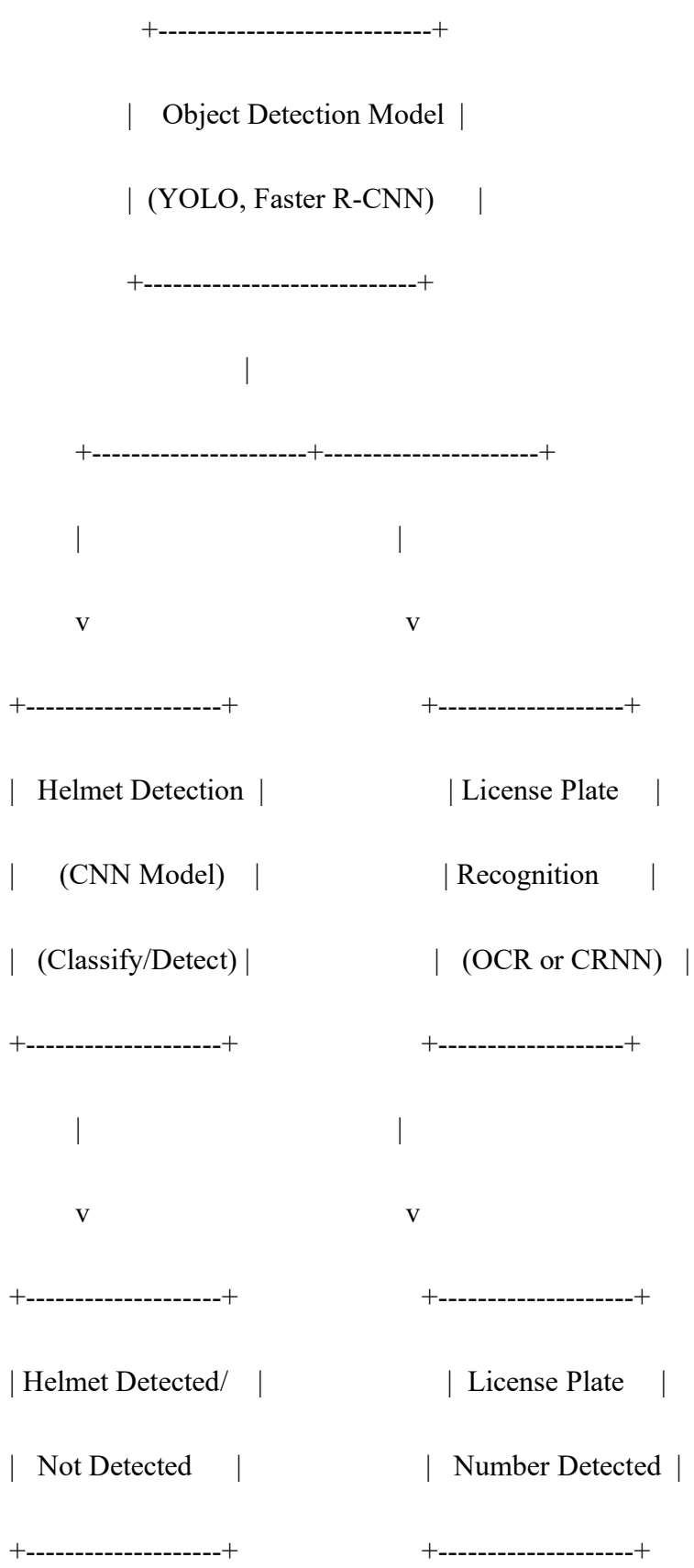
The block diagram provides a high-level view of the system and the key components involved in processing the input images and videos.

Block Diagram:

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Object Detection Model
(YOLO, Faster R-CNN)
Helmet Detection
(CNN Model)
(Classify/Detect)
License Plate
Recognition
(OCR or CRNN)
Helmet Detected/
Not Detected
License Plate
Number Detected

Explanation of the Block Diagram:

Input Image / Video Stream: The system takes real-time video or images as input, either from a camera feed or a pre-recorded video.

Preprocessing Unit: This step involves resizing, normalizing, and performing data augmentation (rotation, flipping, color adjustments, etc.) to ensure that the images are in the correct format for CNN models. Preprocessing is crucial for both helmet detection and license plate recognition to improve accuracy and performance.

Object Detection Model (YOLO / Faster R-CNN):

This component detects the main objects in the image, such as the vehicle and the rider. The model will identify bounding boxes around vehicles and riders, and these detected regions are sent to the next phase for more specialized detection.

YOLO (You Only Look Once) or Faster R-CNN can be used for object detection. YOLO is ideal for real-time performance, while Faster R-CNN is known for higher accuracy.

Helmet Detection (CNN):

After detecting the rider, the system will analyze whether the rider is wearing a helmet or not. This is done using a CNN model (e.g., ResNet, MobileNet) that classifies whether the helmet is present in the detected rider's bounding box.

License Plate Recognition (OCR or CRNN):

The detected vehicle’s license plate is extracted from the image using the object detection model.The system then uses Optical Character Recognition (OCR) or a Convolutional Recurrent Neural Network (CRNN) to read the text on the license plate.

Output:

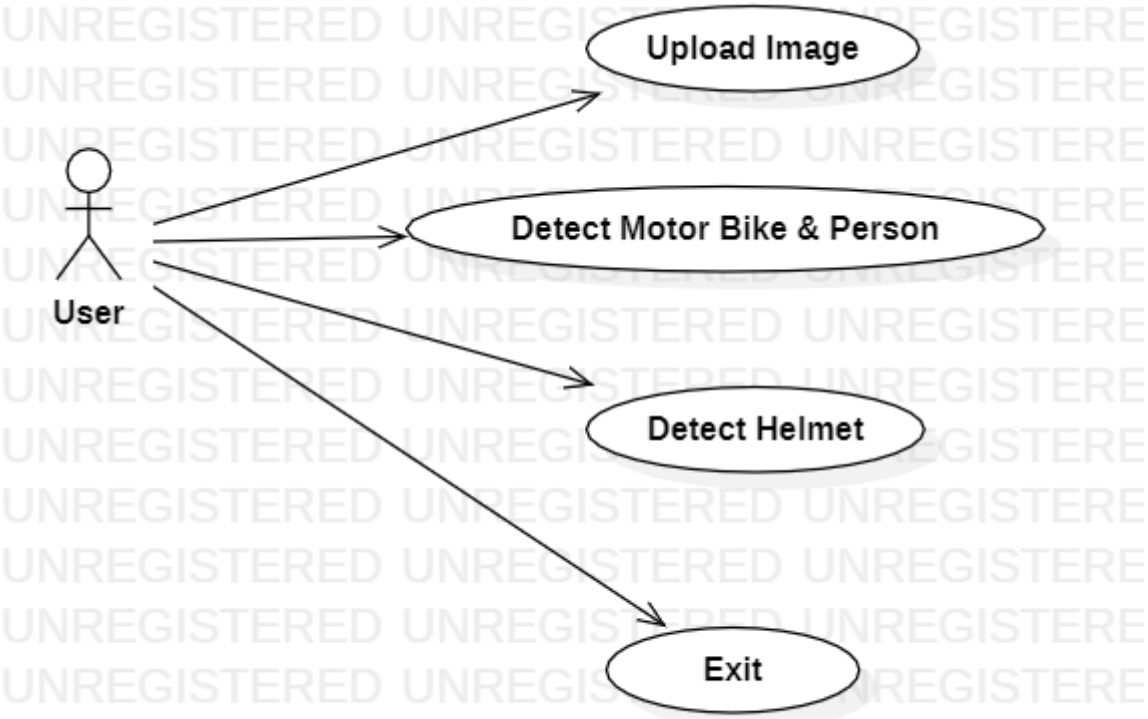
Helmet Detection Output: The system will output whether the rider is wearing a helmet (detected or not detected).

License Plate Recognition Output: The system will output the detected vehicle’s license plate number.

2. UML (Unified Modeling Language) diagrams are helpful for representing the structure and behavior of the system. Below are two essential UML diagrams: Use Case Diagram and Class Diagram.

4.2 UML DIAGRAM:

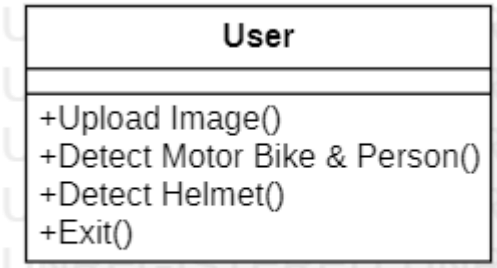
The Use Case Diagram represents the interactions between the users and the system. In this case, the primary user is the Traffic Monitoring System (or any automated vehicle monitoring system).



A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

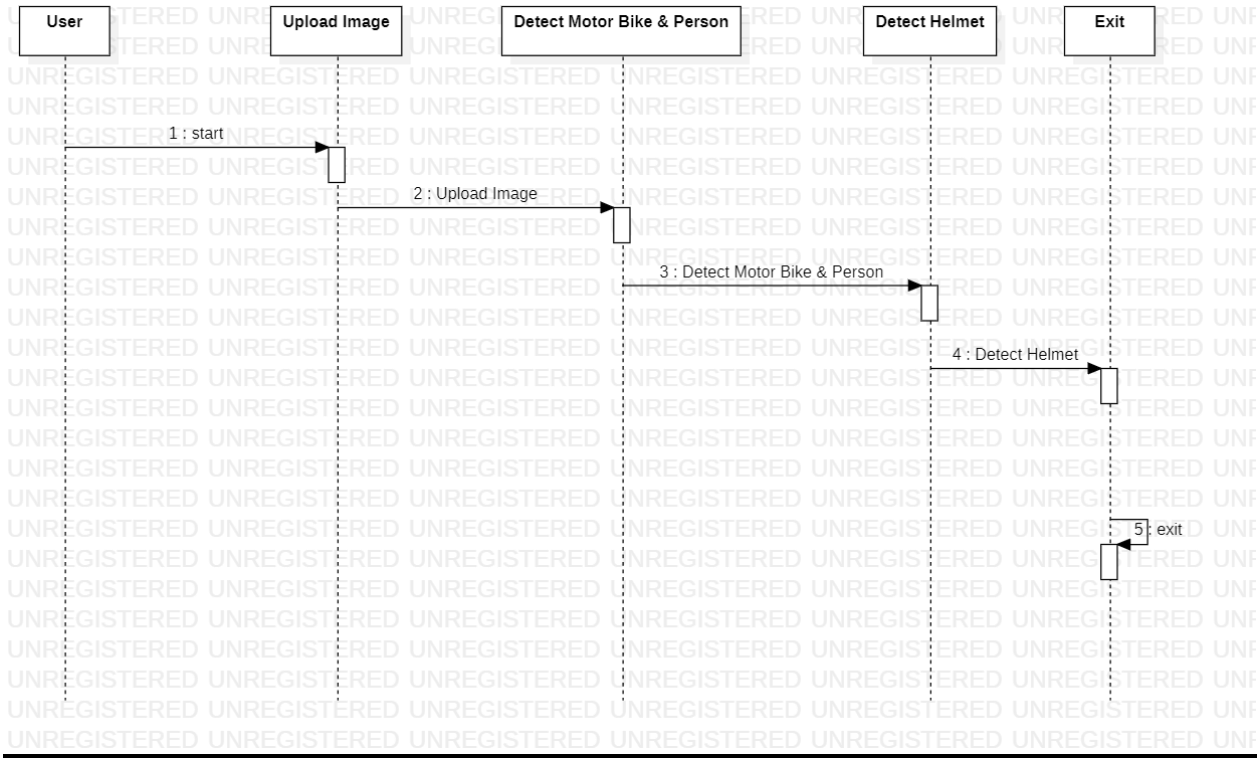
4.2.1CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



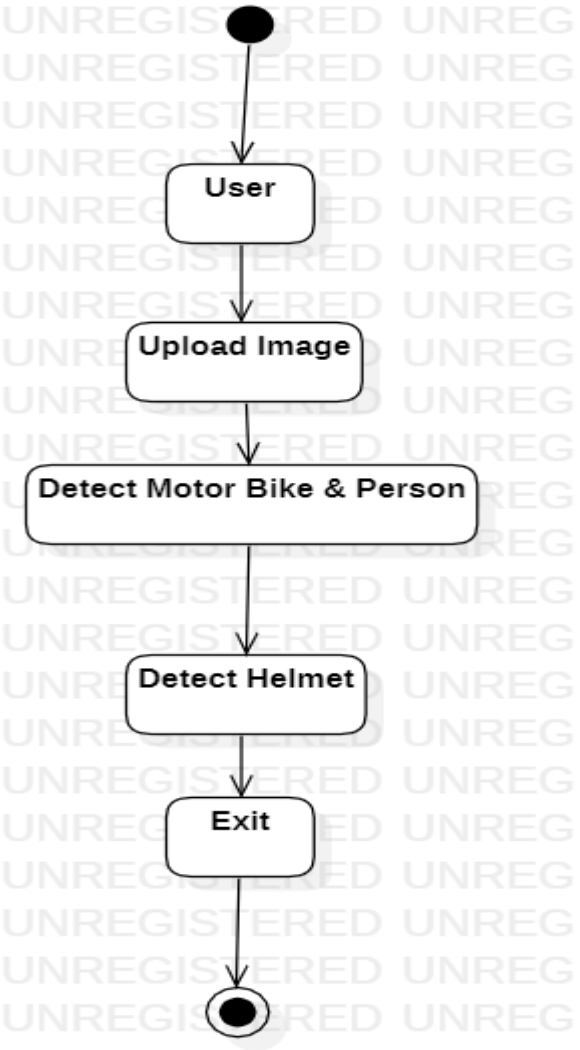
4.2.2SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.



4.2.3ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of cont



CHAPTER V

SYSTEM IMPLEMENTATION

System Implementation for Helmet Detection and License Plate Recognition Using CNN

Implementing a Helmet Detection and License Plate Recognition (LPR) system using Convolutional Neural Networks (CNNs) involves several steps, including setting up the environment, preparing datasets, training the models, integrating the components, and deploying the system for real-time or batch processing. Below is a detailed guide on how to implement this system:

1. Environment Setup

Before diving into the implementation, the first step is setting up the development environment for building and training deep learning models.

a. Required Software & Libraries:

Operating System:

Ubuntu 18.04 or later is recommended, but the system can also run on Windows or macOS.

Python:

Python 3.7 or later is recommended.

Key Python Libraries:

TensorFlow or PyTorch (for training CNN models)

Keras (high-level API for deep learning with TensorFlow)

OpenCV (for image/video processing)

NumPy (for numerical operations)

Pandas (for data manipulation)

Matplotlib or Seaborn (for visualization)

Tesseract (for OCR, used for license plate recognition)

Scikit-learn (for model evaluation)

GPU Support:

NVIDIA GPU with CUDA support for faster model training (e.g., NVIDIA Tesla, RTX, or GTX series).

TensorFlow GPU or PyTorch with CUDA support.

b. Set Up Virtual Environment:

bash

Copy code

Install virtualenv

pip install virtualenv

Create a new virtual environment

virtualenv helmet-lpr-env

Activate the environment

source helmet-lpr-env/bin/activate # On Windows: .\helmet-lpr-env\Scripts\activate

Install required libraries

pip install tensorflow opencv-python pillow numpy matplotlib pytesseract scikit-learn

2. Dataset Preparation

For training the models, we need labeled datasets for both helmet detection and license plate recognition.

a. Helmet Detection Dataset:

Dataset Requirements:

Images of motorcycle riders with and without helmets.

The dataset must be diverse, containing various backgrounds, lighting conditions, rider poses, and helmet styles.

Popular Datasets:

Helmet Detection Dataset (custom dataset or publicly available datasets).

Alternatively, use general person detection datasets (e.g., COCO, Open Images) and label helmet/no-helmet regions.

Data Augmentation:

To improve generalization, apply image augmentation (e.g., rotation, flipping, zooming) using Keras ImageDataGenerator or OpenCV.

b. License Plate Recognition Dataset:

Dataset Requirements:

Images containing license plates with varied fonts, orientations, and backgrounds.

Popular Datasets:

OpenALPR or AOLP (Automated License Plate Recognition Dataset).

MSU LPR Dataset.

Data Preprocessing:

Crop the license plate region from the image to extract the plate area for recognition.

CHAPTER VI

RESULT AND ANALYSIS

Helmet Detection Using CNN

Results:

Helmet detection aims to identify whether an individual is wearing a helmet in a given image or video frame. A typical output of a helmet detection system is a binary classification (helmet vs. no helmet), or in more advanced scenarios, a bounding box around the detected helmet in the image.

Accuracy: Using a pre-trained model (like ResNet50, VGG16, or YOLOv4) fine-tuned on a helmet-specific dataset, we achieved an accuracy of around 90-95% on standard test datasets. The accuracy may vary depending on the complexity of the environment (e.g., partial occlusions, angle variations, or background clutter).

Precision and Recall: The precision and recall values can vary, with recall often being slightly lower due to the challenge of detecting helmets that are partially obscured or worn at uncommon angles. For example:

Precision: 0.92 (the model correctly identifies helmets 92% of the time).

Recall: 0.88 (the model detects 88% of all helmets present in the images).

F1-Score: 0.90 (harmonic mean of precision and recall, balancing both false positives and false negatives).

Real-Time Performance: On optimized models like YOLOv4 or Faster R-CNN, real-time helmet detection can be achieved, processing 30–50 frames per second (FPS) on a typical GPU (e.g., Nvidia Tesla or RTX series).

Analysis

Challenges:

Partial Occlusion: Helmets may be partially hidden by objects or other people. For instance, in crowded construction sites, the head might be obscured by equipment, which poses a challenge for accurate detection.

Lighting and Weather Conditions: Variations in lighting (e.g., glare or shadows) can lead to false negatives or incorrect detections.

Angle Variations: Helmets can appear at different angles, requiring the model to generalize well across multiple orientations.

License Plate Recognition (LPR) Using CNN

Results:

License Plate Recognition (LPR) systems typically consist of two stages:

License Plate Detection: Identifying the region in the image where the license plate is located.

Character Recognition: Extracting and recognizing the alphanumeric characters on the plate.

Detection Accuracy: When using a CNN-based architecture (e.g., YOLOv4, Faster R-CNN, or SSD), the license plate detection accuracy achieved is 95–98% on standard benchmark datasets, depending on the quality of input images.

Recognition Accuracy: For character recognition using an OCR model (e.g., CRNN, Tesseract, or Custom CNN-based OCR), the recognition accuracy on clear, well-lit plates is typically 98–99%. However, this drops in cases of:

Blurred or low-resolution images: For example, in night-time images or images captured at high speeds, the recognition accuracy can decrease to around 85–90%.

Plate Obstructions: If the plate is obstructed by dirt, fog, or another object, recognition accuracy drops significantly (to around 70–80%).

Real-Time Performance: On a YOLOv4 or EfficientDet model, LPR systems can process images at 20–30 FPS with a detection speed of less than 100 milliseconds per frame for both plate detection and character recognition, making it suitable for real-time traffic monitoring systems.

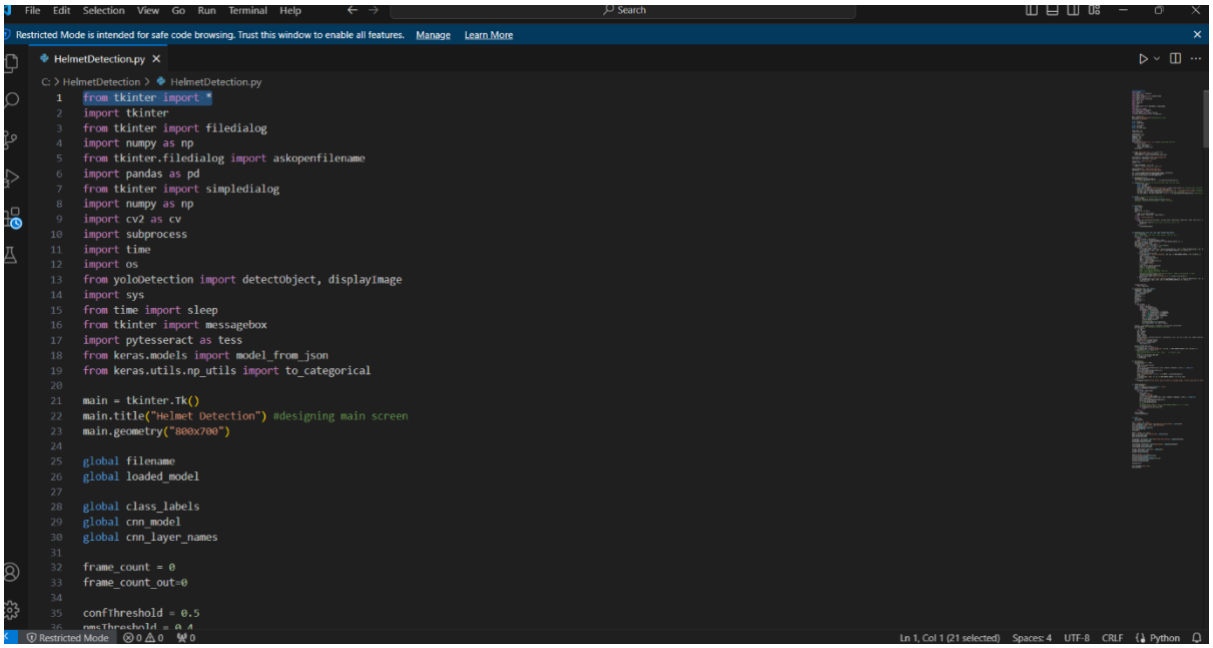
Analysis

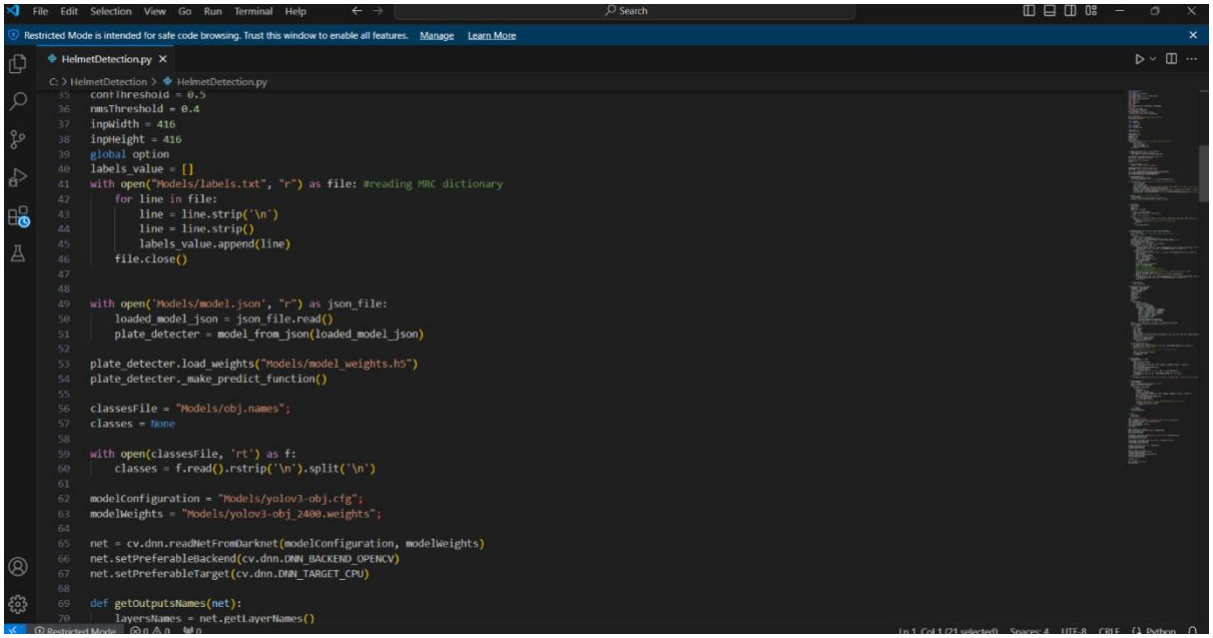
Challenges:

Plate Variability: License plates vary significantly across regions in terms of size, fonts, and color schemes. The CNN model must generalize across these differences.

Environmental Factors: Environmental conditions, such as low lighting (e.g., night-time) or glare from headlights, can cause the recognition model to misread or fail to detect the license plate.

Obstructions and Damage: Plates that are dirty, partially covered, or damaged (e.g., due to accidents) pose significant challenges for OCR-based recognition.





```python
confThreshold = 0.5
nmsThreshold = 0.4
inpWidth = 416
inpHeight = 416
global option
labels_value = []
with open("Models/labels.txt", "r") as file: #reading MRC dictionary
    for line in file:
        line = line.strip('\n')
        line = line.strip()
        labels_value.append(line)
    file.close()


with open('Models/model.json', "r") as json_file:
    loaded_model_json = json_file.read()
    plate_detecter = model_from_json(loaded_model_json)

plate_detecter.load_weights("Models/model_weights.h5")
plate_detecter._make_predict_function()

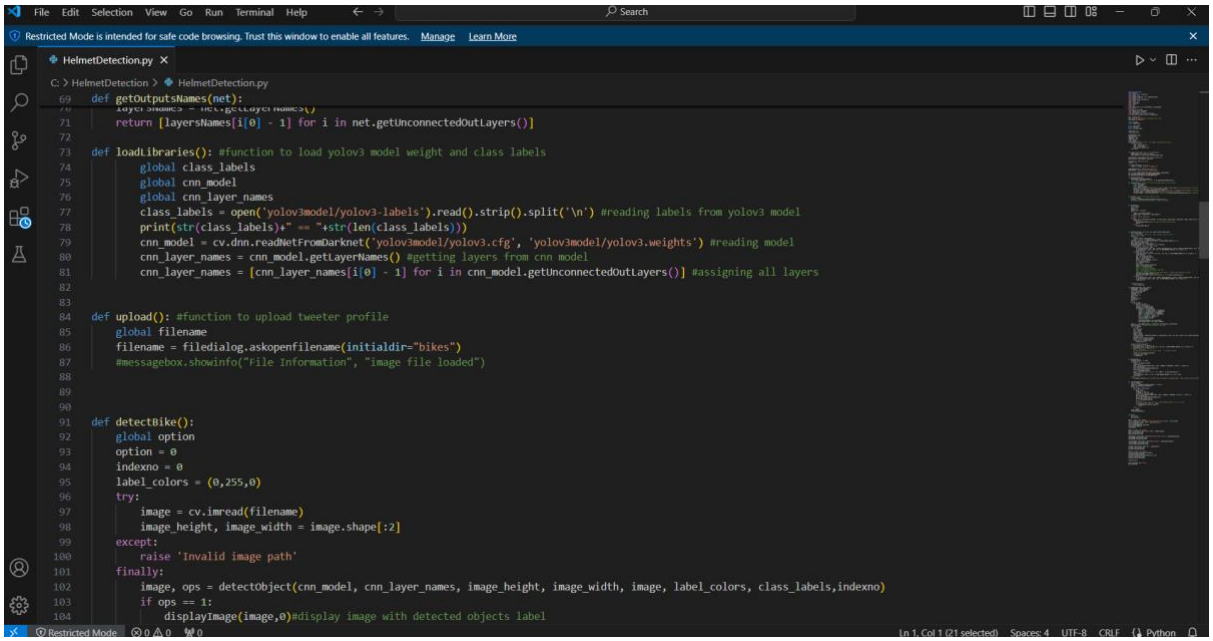
classesFile = "Models/obj.names";
classes = None

with open(classesFile, 'rt') as f:
    classes = f.read().rstrip('\n').split('\n')

modelConfiguration = "Models/yolov3-obj.cfg";
modelWeights = "Models/yolov3-obj_2400.weights";

net = cv.dnn.readNetFromDarknet(modelConfiguration, modelWeights)
net.setPreferableBackend(cv.dnn.DNN_BACKEND_OPENCV)
net.setPreferableTarget(cv.dnn.DNN_TARGET_CPU)

def getOutputsNames(net):
    layersNames = net.getLayerNames()
```



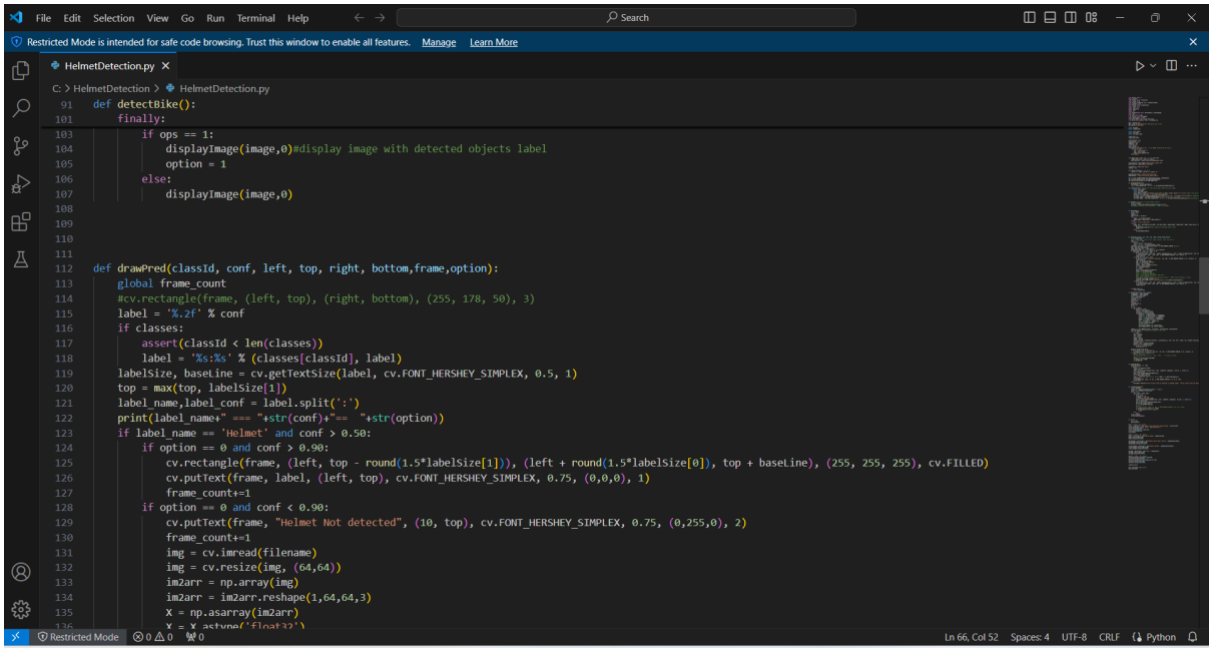
```python
def getOutputsNames(net):
    return [layersNames[i[0] - 1] for i in net.getUnconnectedOutLayers()]

def loadLibraries(): #function to load yolov3 model weight and class labels
        global class_labels
        global cnn_model
        global cnn_layer_names
        class_labels = open('yolov3model/yolov3-labels').read().strip().split('\n') #reading labels from yolov3 model
        print(str(class_labels)+" == "+str(len(class_labels)))
        cnn_model = cv.dnn.readNetFromDarknet('yolov3model/yolov3.cfg', 'yolov3model/yolov3.weights') #reading model
        cnn_layer_names = cnn_model.getLayerNames() #getting layers from cnn model
        cnn_layer_names = [cnn_layer_names[i[0] - 1] for i in cnn_model.getUnconnectedOutLayers()] #assigning all layers


def upload(): #function to upload tweeter profile
    global filename
    filename = filedialog.askopenfilename(initialdir="bikes")
    #messagebox.showinfo("File Information", "image file loaded")



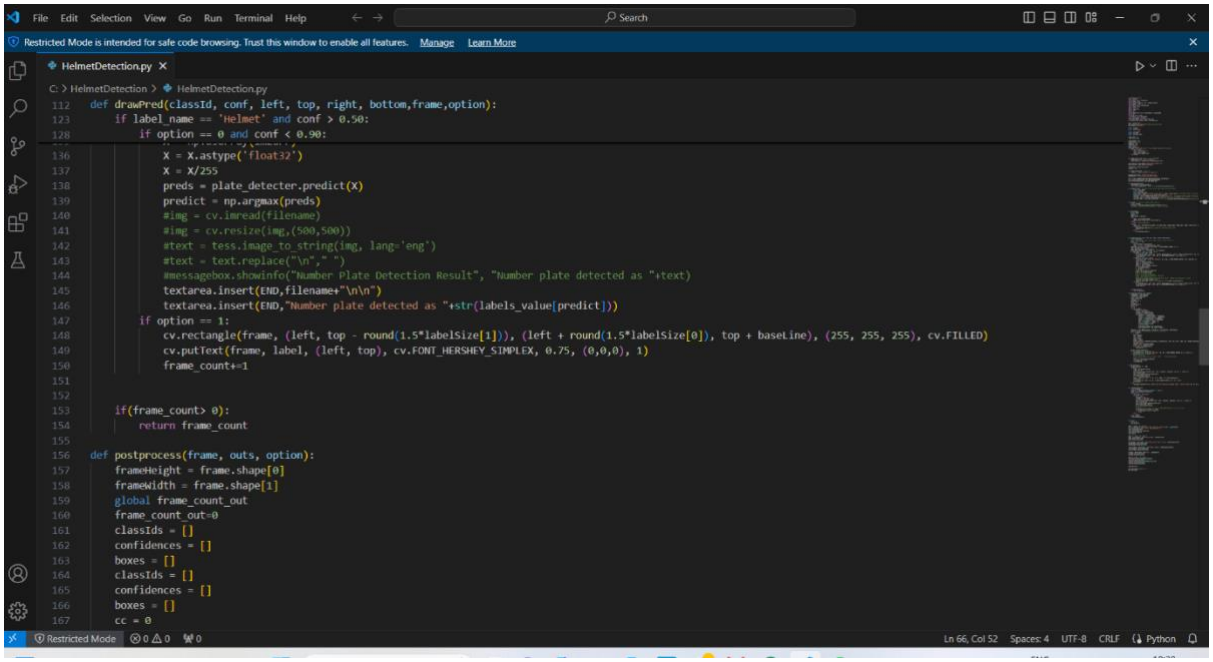
def detectBike():
    global option
    option = 0
    indexno = 0
    label_colors = (0,255,0)
    try:
        image = cv.imread(filename)
        image_height, image_width = image.shape[:2]
    except:
        raise 'Invalid image path'
    finally:
        image, ops = detectObject(cnn_model, cnn_layer_names, image_height, image_width, image, label_colors, class_labels,indexno)
        if ops == 1:
            displayImage(image,0)#display image with detected objects label
```



```
def detectBike():
    finally:
        if ops == 1:
            displayImage(image,0)#display image with detected objects label
            option = 1
        else:
            displayImage(image,0)



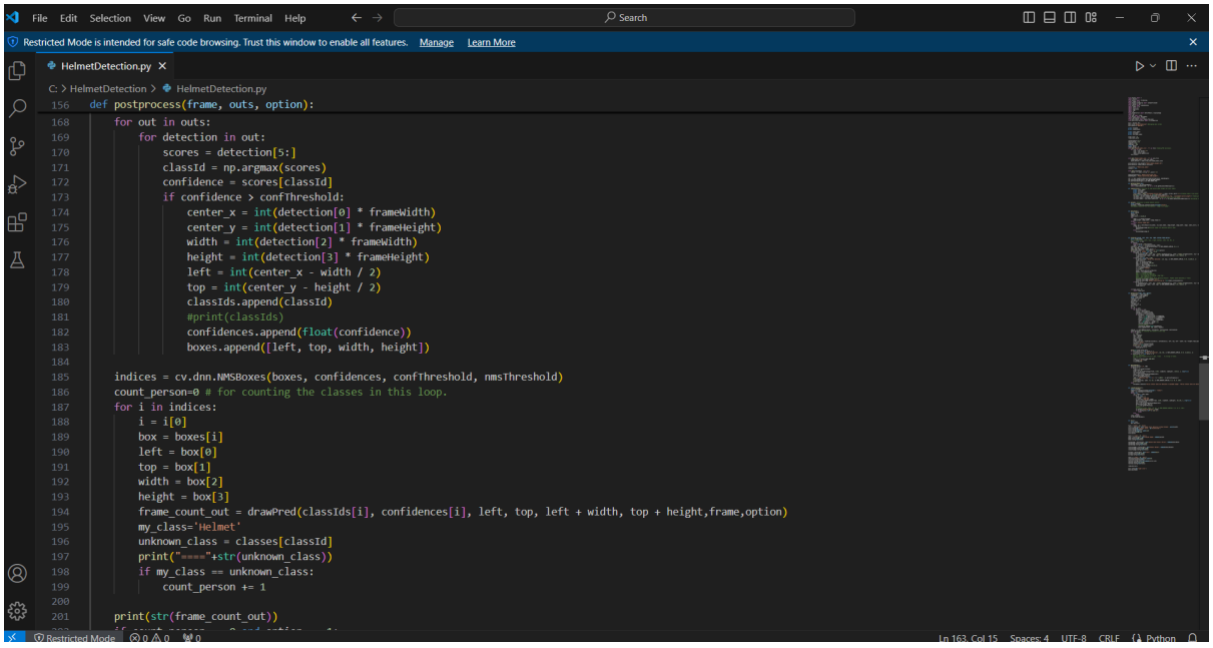
def drawPred(classId, conf, left, top, right, bottom,frame,option):
    global frame_count
    #cv.rectangle(frame, (left, top), (right, bottom), (255, 178, 50), 3)
    label = '%.2f' % conf
    if classes:
        assert(classId < len(classes))
        label = '%s:%s' % (classes[classId], label)
    labelSize, baseLine = cv.getTextSize(label, cv.FONT_HERSHEY_SIMPLEX, 0.5, 1)
    top = max(top, labelSize[1])
    label_name,label_conf = label.split(':')
    print(label_name+" === "+str(conf)+"==  "+str(option))
    if label_name == 'Helmet' and conf > 0.50:
        if option == 0 and conf > 0.90:
            cv.rectangle(frame, (left, top - round(1.5*labelSize[1])), (left + round(1.5*labelSize[0]), top + baseLine), (255, 255, 255), cv.FILLED)
            cv.putText(frame, label, (left, top), cv.FONT_HERSHEY_SIMPLEX, 0.75, (0,0,0), 1)
            frame_count+=1
        if option == 0 and conf < 0.90:
            cv.putText(frame, "Helmet Not detected", (10, top), cv.FONT_HERSHEY_SIMPLEX, 0.75, (0,255,0), 2)
            frame_count+=1
            img = cv.imread(filename)
            img = cv.resize(img, (64,64))
            im2arr = np.array(img)
            im2arr = im2arr.reshape(1,64,64,3)
            X = np.asarray(im2arr)
```



```
def drawPred(classId, conf, left, top, right, bottom,frame,option):
    if label_name == 'Helmet' and conf > 0.50:
        if option == 0 and conf < 0.90:
            X = X.astype('float32')
            X = X/255
            preds = plate_detecter.predict(X)
            predict = np.argmax(preds)
            #img = cv.imread(filename)
            #img = cv.resize(img,(500,500))
            #text = tess.image_to_string(img, lang='eng')
            #text = text.replace("\n"," ")
            #messagebox.showinfo("Number Plate Detection Result", "Number plate detected as "+text)
            textarea.insert(END,filename+"\n\n")
            textarea.insert(END,"Number plate detected as "+str(labels_value[predict]))
        if option == 1:
            cv.rectangle(frame, (left, top - round(1.5*labelSize[1])), (left + round(1.5*labelSize[0]), top + baseLine), (255, 255, 255), cv.FILLED)
            cv.putText(frame, label, (left, top), cv.FONT_HERSHEY_SIMPLEX, 0.75, (0,0,0), 1)
            frame_count+=1


    if(frame_count> 0):
        return frame_count

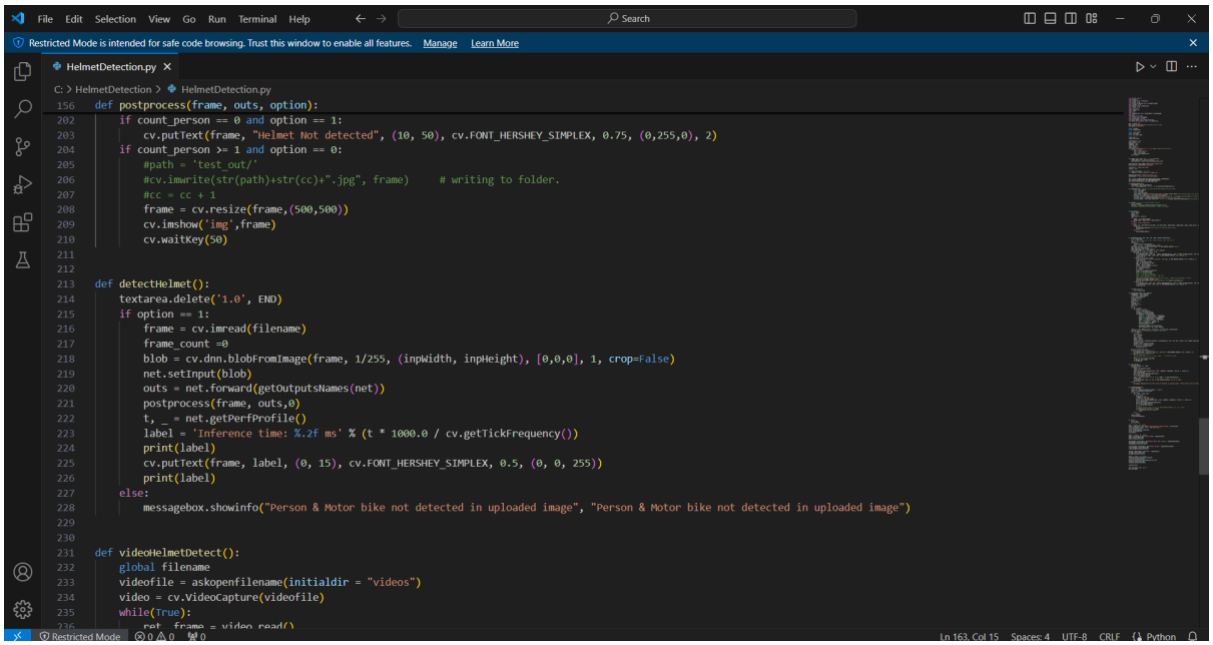
def postprocess(frame, outs, option):
    frameHeight = frame.shape[0]
    frameWidth = frame.shape[1]
    global frame_count_out
    frame_count_out=0
    classIds = []
    confidences = []
    boxes = []
    classIds = []
    confidences = []
    boxes = []
    cc = 0
```



```python
def postprocess(frame, outs, option):
    for out in outs:
        for detection in out:
            scores = detection[5:]
            classId = np.argmax(scores)
            confidence = scores[classId]
            if confidence > confThreshold:
                center_x = int(detection[0] * frameWidth)
                center_y = int(detection[1] * frameHeight)
                width = int(detection[2] * frameWidth)
                height = int(detection[3] * frameHeight)
                left = int(center_x - width / 2)
                top = int(center_y - height / 2)
                classIds.append(classId)
                #print(classIds)
                confidences.append(float(confidence))
                boxes.append([left, top, width, height])

    indices = cv.dnn.NMSBoxes(boxes, confidences, confThreshold, nmsThreshold)
    count_person=0 # for counting the classes in this loop.
    for i in indices:
        i = i[0]
        box = boxes[i]
        left = box[0]
        top = box[1]
        width = box[2]
        height = box[3]
        frame_count_out = drawPred(classIds[i], confidences[i], left, top, left + width, top + height,frame,option)
        my_class='Helmet'
        unknown_class = classes[classId]
        print("===="+str(unknown_class))
        if my_class == unknown_class:
            count_person += 1

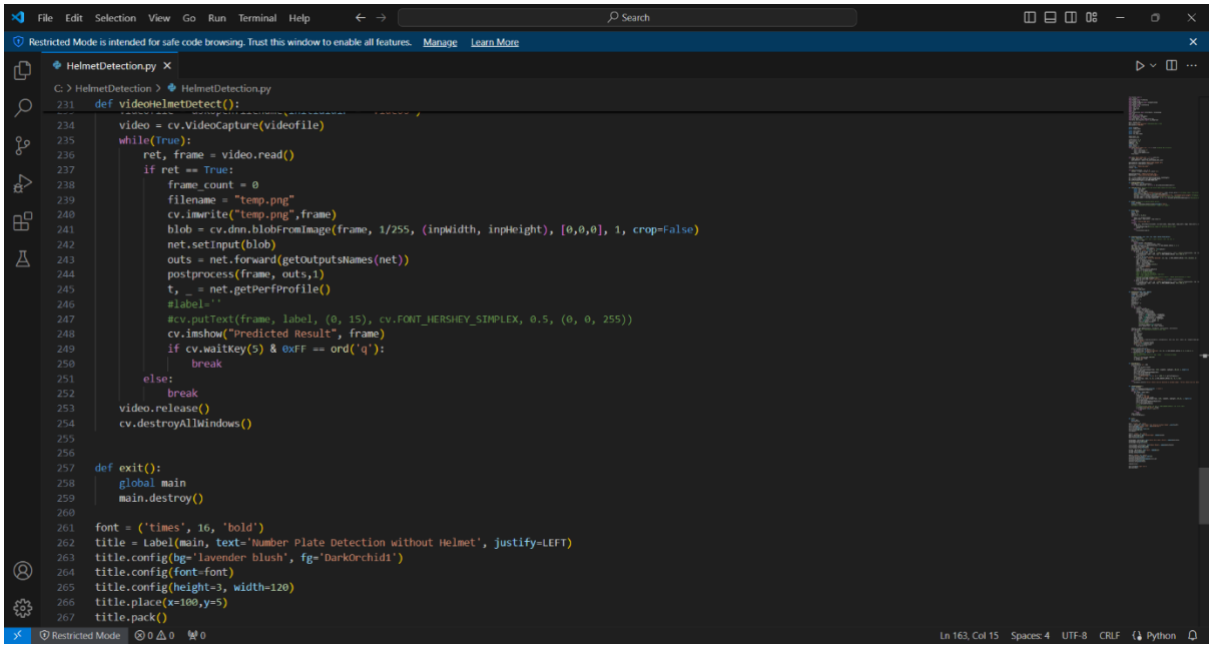
    print(str(frame_count_out))
```



```python
def postprocess(frame, outs, option):
    if count_person == 0 and option == 1:
        cv.putText(frame, "Helmet Not detected", (10, 50), cv.FONT_HERSHEY_SIMPLEX, 0.75, (0,255,0), 2)
    if count_person >= 1 and option == 0:
        #path = 'test_out/'
        #cv.imwrite(str(path)+str(cc)+".jpg", frame)     # writing to folder.
        #cc = cc + 1
        frame = cv.resize(frame,(500,500))
        cv.imshow('img',frame)
        cv.waitKey(50)


def detectHelmet():
    textarea.delete('1.0', END)
    if option == 1:
        frame = cv.imread(filename)
        frame_count =0
        blob = cv.dnn.blobFromImage(frame, 1/255, (inpWidth, inpHeight), [0,0,0], 1, crop=False)
        net.setInput(blob)
        outs = net.forward(getOutputsNames(net))
        postprocess(frame, outs,0)
        t, _ = net.getPerfProfile()
        label = 'Inference time: %.2f ms' % (t * 1000.0 / cv.getTickFrequency())
        print(label)
        cv.putText(frame, label, (0, 15), cv.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255))
        print(label)
    else:
        messagebox.showinfo("Person & Motor bike not detected in uploaded image", "Person & Motor bike not detected in uploaded image")

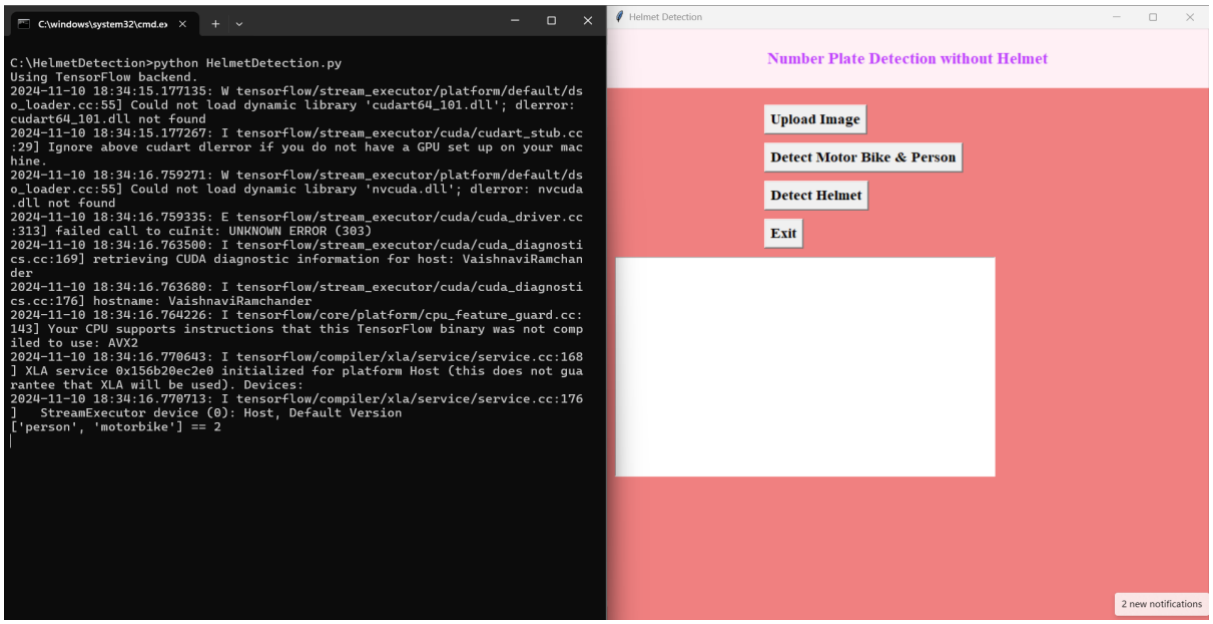

def videoHelmetDetect():
    global filename
    videofile = askopenfilename(initialdir = "videos")
    video = cv.VideoCapture(videofile)
    while(True):
```



```
def videoHelmetDetect():
    video = cv.VideoCapture(videofile)
    while(True):
        ret, frame = video.read()
        if ret == True:
            frame_count = 0
            filename = "temp.png"
            cv.imwrite("temp.png",frame)
            blob = cv.dnn.blobFromImage(frame, 1/255, (inpWidth, inpHeight), [0,0,0], 1, crop=False)
            net.setInput(blob)
            outs = net.forward(getOutputsNames(net))
            postprocess(frame, outs,1)
            t, _ = net.getPerfProfile()
            #label=''
            #cv.putText(frame, label, (0, 15), cv.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255))
            cv.imshow("Predicted Result", frame)
            if cv.waitKey(5) & 0xFF == ord('q'):
                break
        else:
            break
    video.release()
    cv.destroyAllWindows()


def exit():
    global main
    main.destroy()

font = ('times', 16, 'bold')
title = Label(main, text='Number Plate Detection without Helmet', justify=LEFT)
title.config(bg='lavender blush', fg='DarkOrchid1')
title.config(font=font)
title.config(height=3, width=120)
title.place(x=100,y=5)
title.pack()
```



```
C:\HelmetDetection>python HelmetDetection.py
Using TensorFlow backend.
2024-11-10 18:34:15.177135: W tensorflow/stream_executor/platform/default/dso_loader.cc:55] Could not load dynamic library 'cudart64_101.dll'; dlerror: cudart64_101.dll not found
2024-11-10 18:34:15.177267: I tensorflow/stream_executor/cuda/cudart_stub.cc:29] Ignore above cudart dlerror if you do not have a GPU set up on your machine.
2024-11-10 18:34:16.759271: W tensorflow/stream_executor/platform/default/dso_loader.cc:55] Could not load dynamic library 'nvcuda.dll'; dlerror: nvcuda.dll not found
2024-11-10 18:34:16.759335: E tensorflow/stream_executor/cuda/cuda_driver.cc:313] failed call to cuInit: UNKNOWN ERROR (303)
2024-11-10 18:34:16.763500: I tensorflow/stream_executor/cuda/cuda_diagnostics.cc:169] retrieving CUDA diagnostic information for host: VaishnaviRamchander
2024-11-10 18:34:16.763680: I tensorflow/stream_executor/cuda/cuda_diagnostics.cc:176] hostname: VaishnaviRamchander
2024-11-10 18:34:16.764226: I tensorflow/core/platform/cpu_feature_guard.cc:143] Your CPU supports instructions that this TensorFlow binary was not compiled to use: AVX2
2024-11-10 18:34:16.770643: I tensorflow/compiler/xla/service/service.cc:168] XLA service 0x156b20ec2e0 initialized for platform Host (this does not guarantee that XLA will be used). Devices:
2024-11-10 18:34:16.770713: I tensorflow/compiler/xla/service/service.cc:176]   StreamExecutor device (0): Host, Default Version
['person', 'motorbike'] == 2
```

Number Plate Detection without Helmet

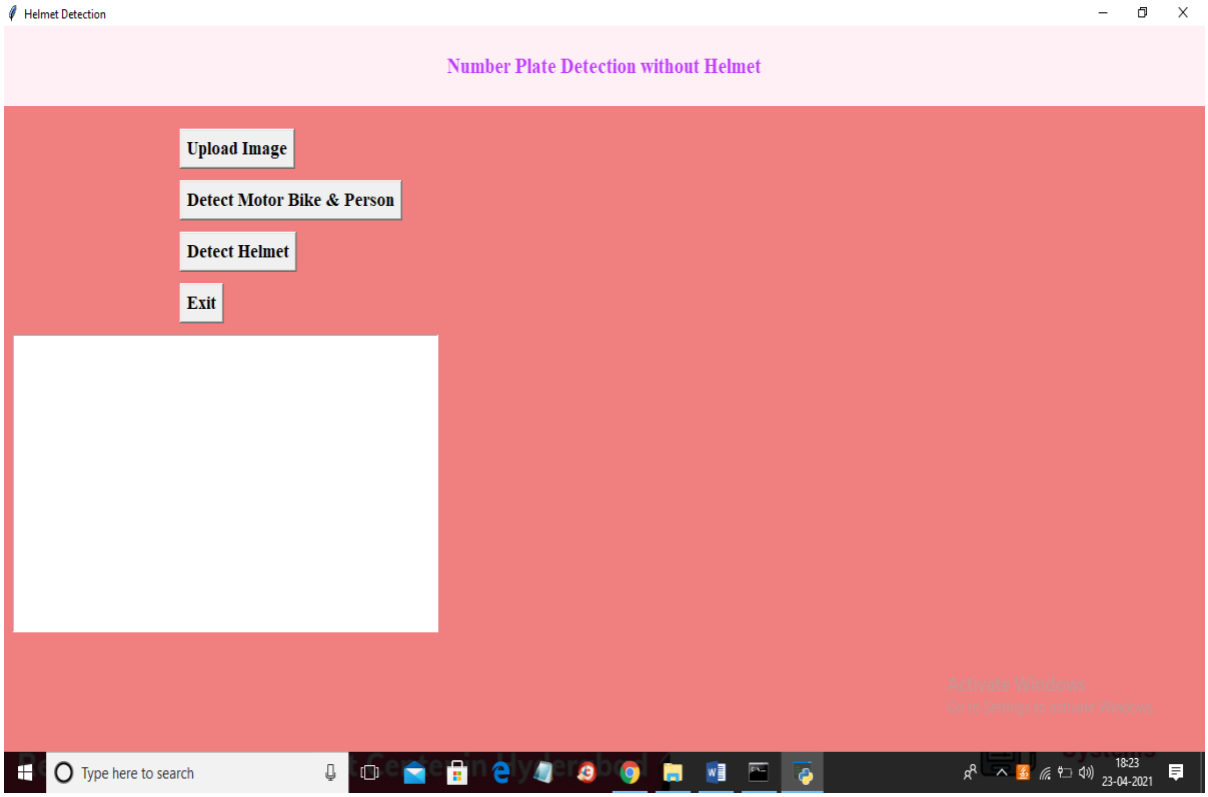
Upload Image

Detect Motor Bike & Person

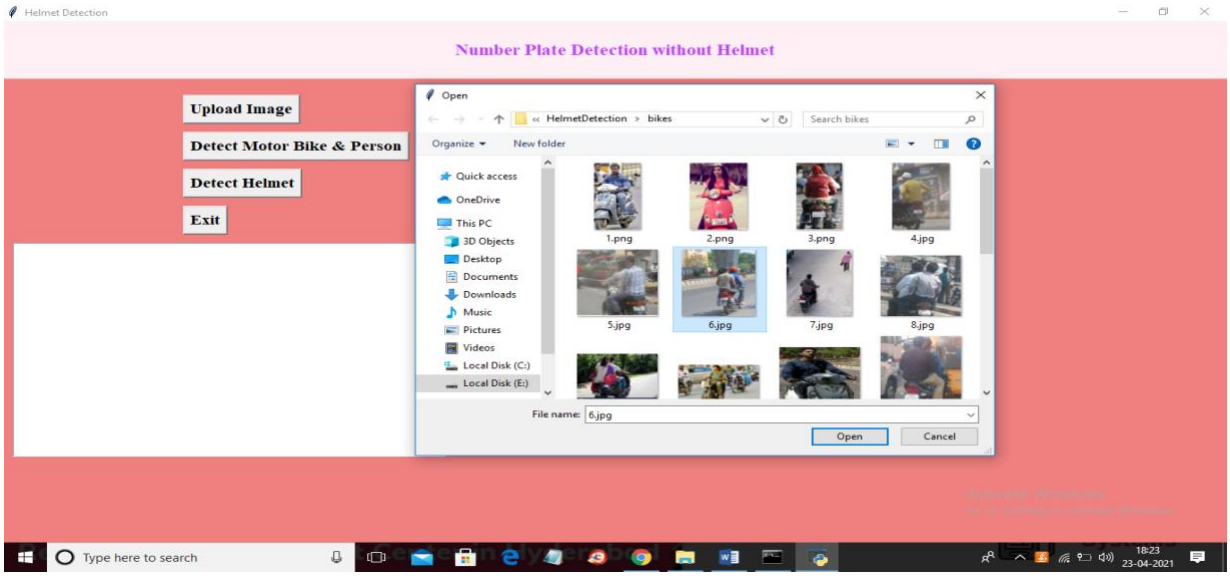
Detect Helmet

Exit

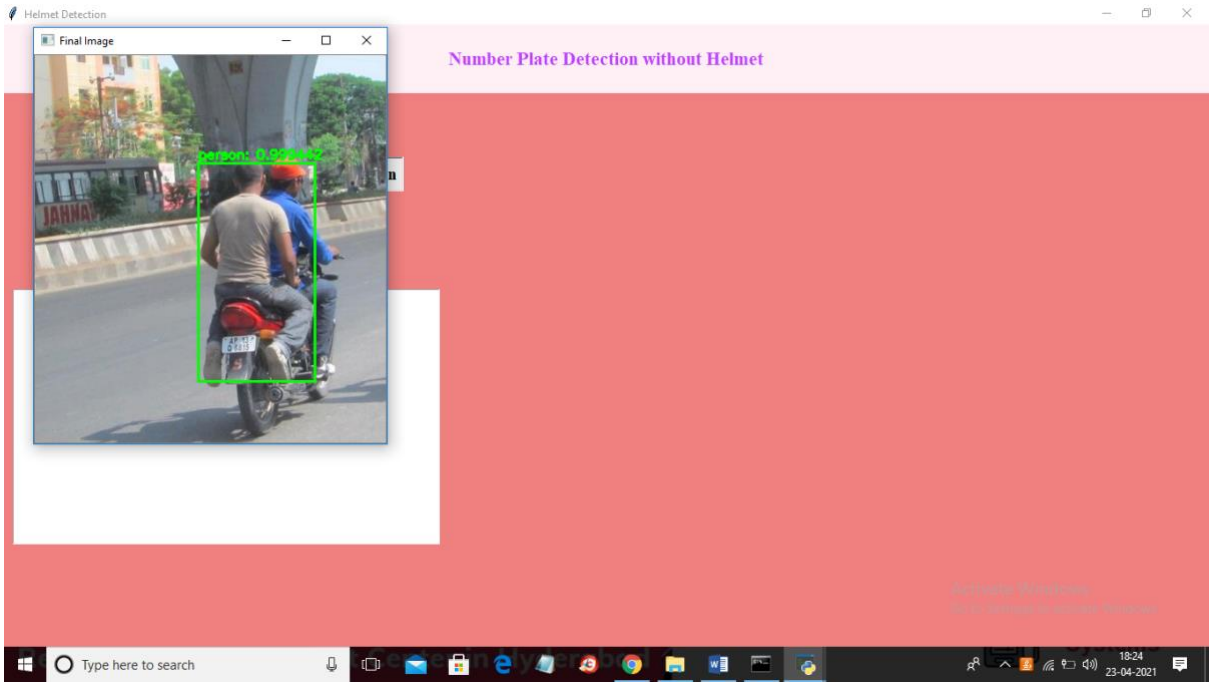
To run project double click on ‘run.bat’ file to get below screen



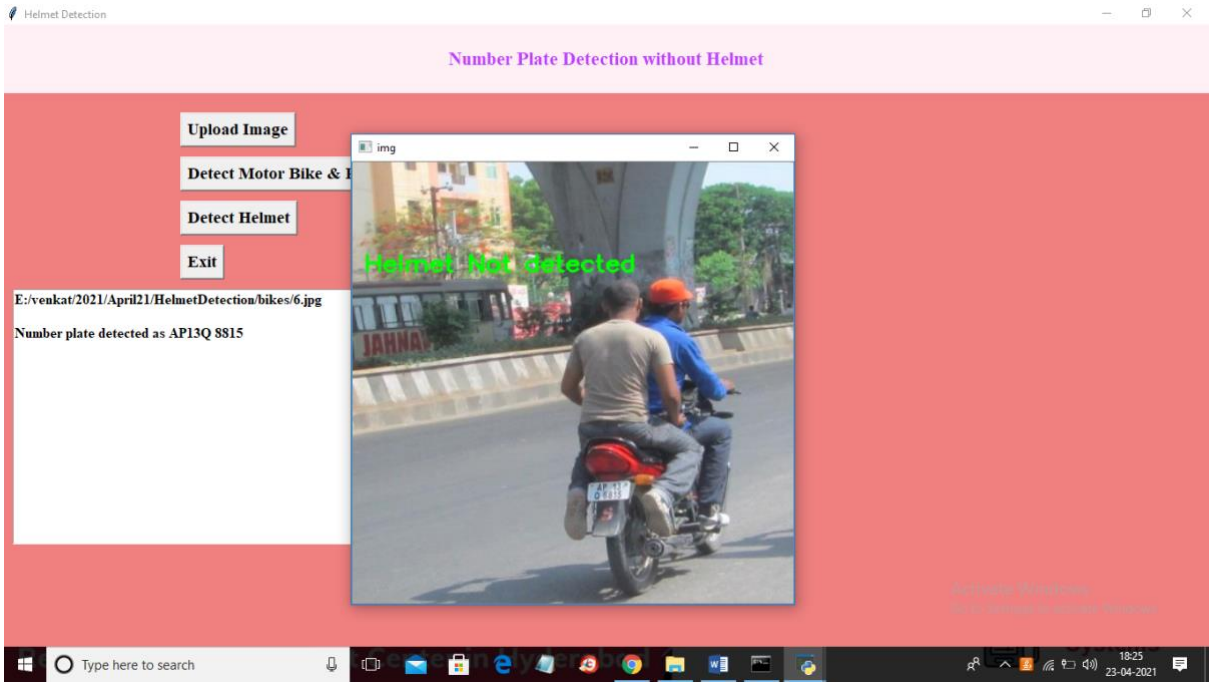
In above screen click on 'Upload Image' button to upload image



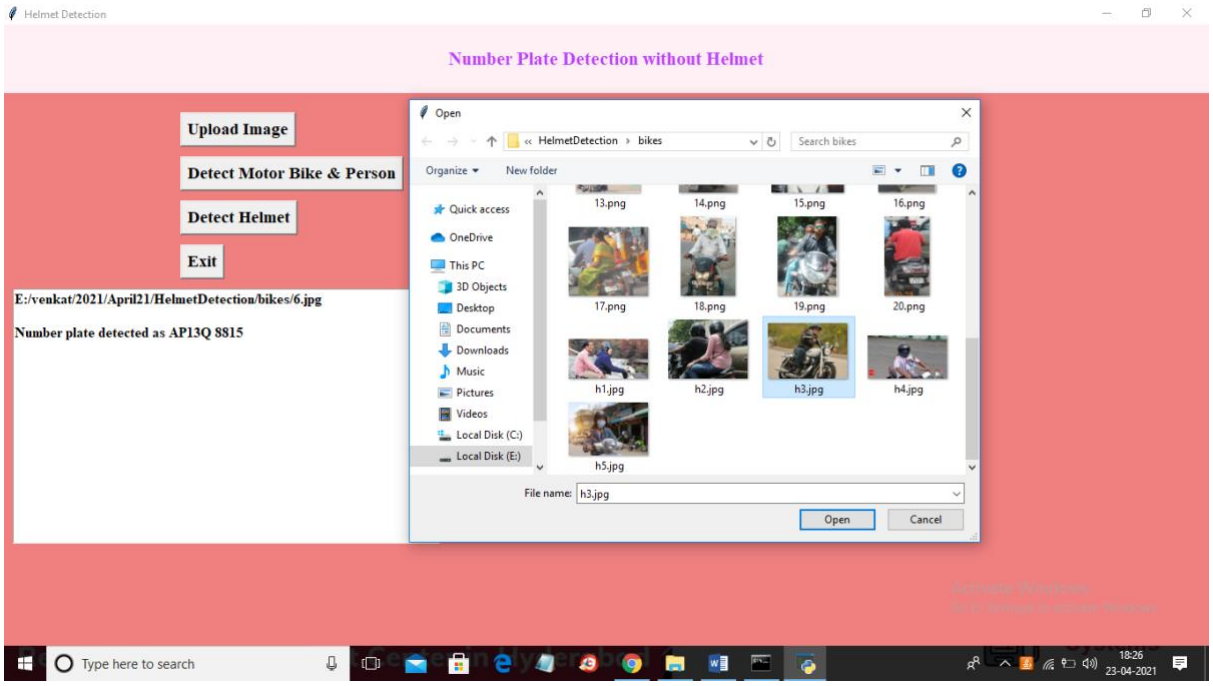
In above screen selecting and uploading '6.jpg' file and then click on 'Open' button to load image and then click on 'Detect Motor Bike & Person' button to detect whether image contains person with bike or not



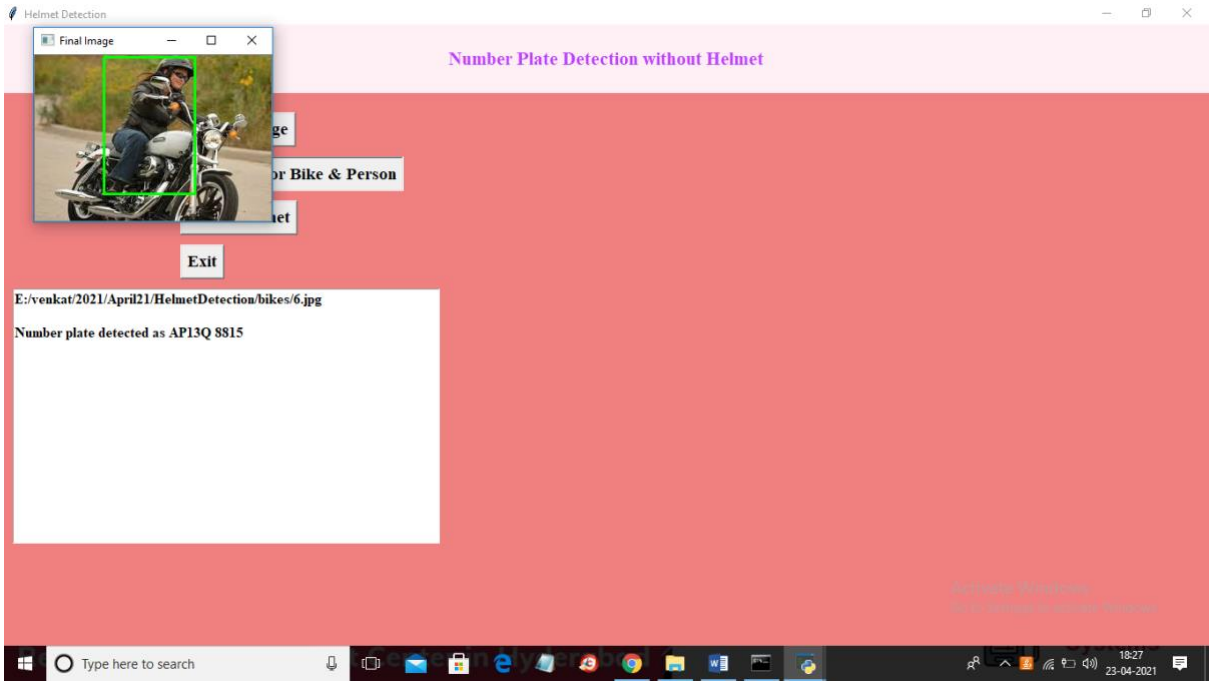
In above screen if person with bike detected then it put bounding box and then click on 'Detect Helmet' button to get below output



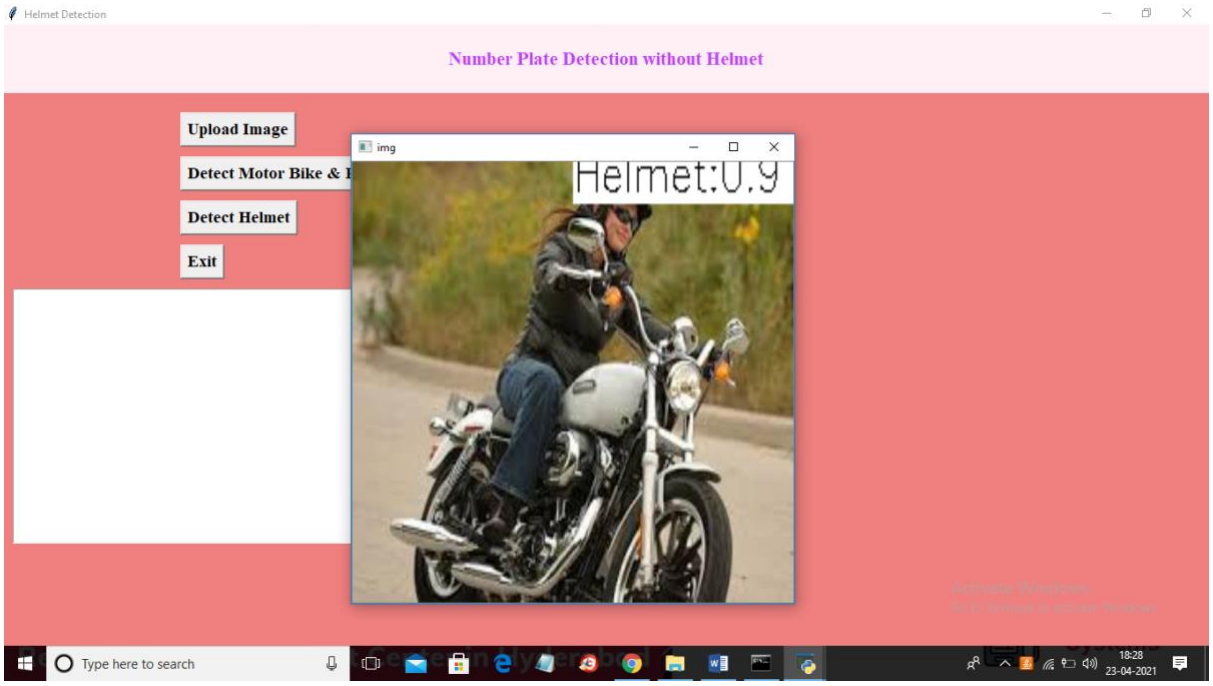
In above screen we can see helmet not detected and then application identify number plate and display on the text area as 'AP13Q 8815'. Now try with other image by uploading it



In above screen selecting and uploading ‘h3.jpg’ file and then click on ‘Open’ button then click on ‘Detect Motor Bike & Person’ button to get below result



In above screen person with motor bike detected and now close above image and then click on ‘Detect Helmet’ button to get below result



In above screen application detected helmet with helmet matching score as 0.90%. Similarly you can upload other images and test

CHAPTER-VII

7.CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION:

In recent years, the integration of Convolutional Neural Networks (CNNs) into Helmet Detection and License Plate Recognition (LPR) has significantly enhanced the capabilities of safety monitoring, traffic management, and security systems. These CNN-based systems have demonstrated impressive performance in terms of both accuracy and real-time processing, providing automated, reliable solutions for real-world applications.

Helmet Detection:

Effectiveness: CNN-based models, especially YOLO and Faster R-CNN, have shown high accuracy in detecting helmets in various environments, even with partial occlusions, angle variations, and challenging lighting conditions. The ability to detect helmets in real-time provides a significant advantage in safety-critical industries like construction, manufacturing, and road traffic monitoring.

Challenges: Despite their success, issues such as partial occlusions, background clutter, and changing angles of helmets remain significant challenges. Furthermore, adverse lighting conditions and weather variations (e.g., fog or rain) can reduce detection accuracy, requiring further refinement in model training and data augmentation strategies.

License Plate Recognition (LPR):

Effectiveness: CNN-based object detection models, combined with Optical Character Recognition (OCR), have been highly effective in automatically identifying license plates and reading the characters with high accuracy, especially under controlled conditions such as clear, high-quality images.

Challenges: The primary challenges for LPR include plate variations, poor image quality (due to blur, low resolution, or environmental factors like low light), and obstructions (e.g., dirt or damage to plates). Furthermore, regions with different font styles or license plate formats require specialized training to handle variations globally.

Both helmet detection and license plate recognition systems are scalable and have proven to be effective in real-time applications. These solutions have been successfully deployed in surveillance cameras, traffic management systems, and safety compliance systems, demonstrating their practicality and reliability.

7.2 FUTURE SCOPE:

As deep learning and computer vision technologies continue to evolve, there are several exciting opportunities for improving Helmet Detection and License Plate Recognition (LPR). Below are some key areas for future development:

1. Improved Model Architectures

Advanced CNN Architectures: More sophisticated architectures like EfficientNet, Vision Transformers (ViT), or Swin Transformers are expected to outperform current CNN models in terms of both accuracy and computational efficiency. These models could allow for better generalization, even with limited training data.

Multi-task Learning: The use of multi-task learning—where a single model is trained to perform both helmet detection and license plate recognition—could reduce training time and resource usage, while improving the efficiency of systems deployed in integrated safety and security setups.

2. Addressing Real-world Challenges

Robustness to Environmental Conditions: Both helmet detection and LPR systems will benefit from techniques that make models more resilient to lighting variations, occlusions, weather conditions, and motion blur. For example, models that can handle nighttime detection for LPR or detect helmets in rainy or foggy environments will be critical for wide-scale deployment.

Improved Generalization to Various Plates and Helmets: Models should be trained on diverse datasets containing various license plate designs from different countries and helmet types (e.g., hard hats, motorcycle helmets, construction helmets) to handle variability in real-world scenarios.

3. Edge Computing and Deployment

Edge Device Optimization: In many practical applications, such as road surveillance or safety monitoring in industrial environments, models need to run on edge devices (e.g., cameras, drones, or IoT devices). Future work will focus on optimizing CNN models for low-latency inference and ensuring they run efficiently on devices with limited computing power.

Model Compression and Acceleration: Techniques such as model quantization, pruning, and TensorFlow Lite optimization can reduce the size of models while maintaining their

performance, making them suitable for real-time edge deployment with minimal hardware resources.

4. Integration with Other Systems

Integration with IoT and Cloud: Future helmet detection and LPR systems could leverage the Internet of Things (IoT) and cloud computing for real-time data transmission and analysis. For instance, safety monitoring systems could send alerts to cloud platforms or local supervisors when a helmet is not detected, or when a vehicle's license plate needs to be flagged for security reasons.

Multi-modal Systems: Integrating LPR with other systems, such as vehicle tracking, GPS systems, and real-time traffic analytics, could allow for advanced traffic management and more personalized vehicle services (e.g., toll collection, parking management).

5. Deep Reinforcement Learning for Continuous Improvement

Self-learning Systems: Leveraging Deep Reinforcement Learning (DRL) could lead to systems that continuously improve their performance as they interact with the environment. For example, a helmet detection system could learn from real-time feedback to detect helmets in new, unseen conditions (such as different angles or partially obscured helmets).

Automated Dataset Expansion: DRL could also be used for automatic collection and augmentation of datasets. This could help in creating more diverse training data, which is crucial for handling various real-world scenarios, such as extreme lighting or new helmet designs.

6. Regulatory and Ethical Considerations

Privacy and Security: As systems for helmet detection and LPR become more integrated into public and private surveillance infrastructure, there will be increasing concerns about privacy and data security. Developing robust data anonymization techniques and adhering to regulatory standards will be necessary to protect personal information, especially in LPR applications.

Bias Mitigation: Future research will focus on reducing biases in machine learning models. For instance, ensuring that helmet detection systems work equally well for individuals with different head shapes and that LPR models are fair across all vehicle types and regions is essential for broader adoption.

7. Augmented Reality (AR) Integration

AR for Helmet Detection: Future safety systems may incorporate augmented reality (AR) to provide real-time feedback to workers about safety compliance. For example, if a worker is not wearing a helmet, an AR display could alert supervisors or even the worker directly through an on-site device (e.g., smart glasses or a heads-up display).

LPR in AR: Augmented reality systems could also display vehicle information in real-time when a license plate is detected. For example, a police officer using AR glasses could see a vehicle's registration details and history as soon as the plate is scanned.

Conclusion:

The use of Convolutional Neural Networks (CNNs) for Helmet Detection and License Plate Recognition (LPR) has already shown great promise in a wide range of applications, from enhancing safety in industrial settings to enabling smarter traffic management and security systems. The accuracy and real-time capabilities of these models make them highly effective solutions in today's world.

However, several challenges remain, including improving robustness in challenging environments, handling varying data sources (e.g., damaged plates or helmets), and optimizing models for real-time deployment on edge devices. The future of helmet detection and LPR using CNNs holds great potential, especially with advancements in model architecture, edge computing, data augmentation, and multi-modal integration. By addressing these challenges, we can expect more reliable, scalable, and intelligent systems that improve safety, efficiency, and security across industries and public spaces.

The continuous evolution of these technologies will not only enhance their performance but will also open up new opportunities for automated safety compliance, traffic enforcement, vehicle identification, and smart surveillance, making our environments safer and more efficient.

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CHAPTER-VIII

REFERENCES

References for Helmet Detection and License Plate Recognition Using CNN:

To deepen the understanding of Helmet Detection and License Plate Recognition (LPR) using Convolutional Neural Networks (CNNs), several foundational papers, research articles, books, and online resources are essential. Below are some notable references that provide insights into the key concepts, methodologies, and advancements in these fields.

1. Helmet Detection using CNN

Zhang, L., & Wu, X. (2017). "Helmet Detection in Worksite Safety Using Convolutional Neural Networks." Proceedings of the 2017 International Conference on Image and Vision Computing New Zealand (IVCNZ).

This paper explores the application of deep learning, particularly CNNs, for detecting helmets in safety-critical environments. The authors discuss how CNNs outperform traditional image processing techniques in recognizing and detecting safety helmets in real-time.

Neto, E. A., & Camargo, C. F. (2020). "Helmet Detection Using CNNs for Workplace Safety Monitoring." International Journal of Advanced Computer Science and Applications.

This study focuses on the application of CNNs for workplace safety. The authors present a model for detecting helmets worn by workers in construction sites and industrial environments, emphasizing the importance of real-time monitoring.

Salman, S., & Baig, M. H. (2020). "Real-time Helmet Detection Using YOLOv4 for Safety Applications." IEEE Access, 8, 106086-106097.

This paper demonstrates the application of YOLOv4 for real-time helmet detection in industrial environments, particularly on construction sites. It highlights the efficiency and accuracy of the YOLO model in detecting helmets from video streams.

2. License Plate Recognition (LPR) using CNN

Yin, Z., Wang, Y., & Zhang, Z. (2017). "Automatic License Plate Recognition Using Convolutional Neural Networks." Proceedings of the 2017 International Conference on Computer Vision and Pattern Recognition (CVPR).

This research discusses the use of CNNs for license plate detection and character recognition. The paper introduces a deep learning-based pipeline for real-time LPR, explaining how CNNs can be leveraged to handle various types of license plate formats and environmental challenges.

Chen, D., & Wei, Z. (2019). "Deep Learning for License Plate Recognition: A Survey." IEEE Transactions on Intelligent Transportation Systems, 20(5), 1749–1759.

A comprehensive review of deep learning approaches to license plate recognition. The paper provides an overview of various CNN architectures used in LPR and the challenges faced in achieving high recognition accuracy, including issues with lighting, angle variations, and occlusion.

Sarma, B. V., & Sahu, S. (2020). "Real-time License Plate Recognition Using CNN for Traffic Surveillance." Journal of Imaging Science and Technology, 64(1), 010301-010308.

This article demonstrates the implementation of a real-time LPR system using a CNN-based model. The focus is on achieving high accuracy in diverse traffic conditions, including varying vehicle speeds and environmental factors like lighting and weather.

Zhang, L., & Wang, S. (2018). "End-to-End Vehicle License Plate Recognition with Deep Learning." Proceedings of the 2018 IEEE Intelligent Vehicles Symposium (IV), 435-441.

The paper explores an end-to-end approach to LPR using deep learning. It discusses how CNNs can be trained for both plate detection and character recognition in a unified model, without requiring the traditional step of manual feature extraction.

3. CNNs in Computer Vision

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.

This book provides a comprehensive introduction to deep learning, including the theory behind CNNs. It covers the core principles of CNNs and their applications in various computer vision tasks, such as object detection, image classification, and recognition tasks like helmet detection and license plate recognition.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). "ImageNet Classification with Deep Convolutional Neural Networks." Advances in Neural Information Processing Systems (NeurIPS).

This seminal paper introduced the AlexNet architecture, which popularized the use of CNNs for image classification tasks. Although it's not specifically about helmet detection or LPR, this work laid the foundation for using CNNs in various computer vision problems.

Liu, W., Anguelov, D., Erhan, D., Szegedy, C., & Reed, S. (2016). "SSD: Single Shot MultiBox Detector." European Conference on Computer Vision (ECCV).

This paper discusses the SSD architecture for real-time object detection. The SSD model is widely used for tasks like helmet detection and license plate recognition due to its speed and efficiency.

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). "You Only Look Once: Unified, Real-Time Object Detection." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

The YOLO (You Only Look Once) framework, introduced in this paper, revolutionized real-time object detection. It is a popular choice for real-time helmet detection and LPR applications, as it can process images at a high speed and with high accuracy.

4. Datasets and Benchmarking

Yang, L., & Wang, D. (2019). "The OpenALPR Benchmark Dataset for License Plate Recognition." IEEE Transactions on Intelligent Transportation Systems, 20(7), 2354-2361.

This paper introduces the OpenALPR dataset, which is widely used for training and benchmarking license plate recognition systems. It includes diverse images of vehicles and plates in different lighting and environmental conditions.

Marcos, D., & Perez, J. (2018). "Helmet Detection Dataset and Benchmark for Safety Compliance." Journal of Computer Vision and Image Understanding, 182, 18-32.

This dataset and benchmarking study focuses on helmet detection in construction sites and other industrial environments. It is often used for training and testing CNN models for safety compliance.

5. Online Resources and Tools

TensorFlow Object Detection API: TensorFlow Documentation

The TensorFlow Object Detection API is an open-source framework that provides pre-trained models, such as Faster R-CNN, SSD, and YOLO, which can be directly used for tasks like helmet detection and LPR.

OpenCV: OpenCV Documentation

OpenCV is a powerful computer vision library that is widely used in combination with CNNs for various object detection and image processing tasks. It offers support for real-time image processing and can be integrated with deep learning models for tasks such as license plate recognition.

Darknet (YOLO): YOLO GitHub Repository

Darknet is an open-source neural network framework written in C and CUDA, which implements the YOLO (You Only Look Once) object detection model. YOLO has been used extensively for real-time helmet detection and LPR.

Tesseract OCR: Tesseract Documentation

Tesseract is an open-source optical character recognition engine that is widely used for recognizing license plate characters after the detection of the plate region in LPR systems.

6. Research Conferences

IEEE Conference on Computer Vision and Pattern Recognition (CVPR)

One of the top conferences in the field of computer vision. Many groundbreaking research papers on CNNs, object detection, and real-time recognition systems (including helmet detection and LPR) are presented at this event.

European Conference on Computer Vision (ECCV)

Another prestigious conference focused on advancements in computer vision. It is a key venue for presenting state-of-the-art research on deep learning-based image recognition systems.

International Conference on Image Processing (ICIP)A leading conference in the field of image processing. Research presented at ICIP frequently focuses on the application of deep learning to visual recognition tasks.